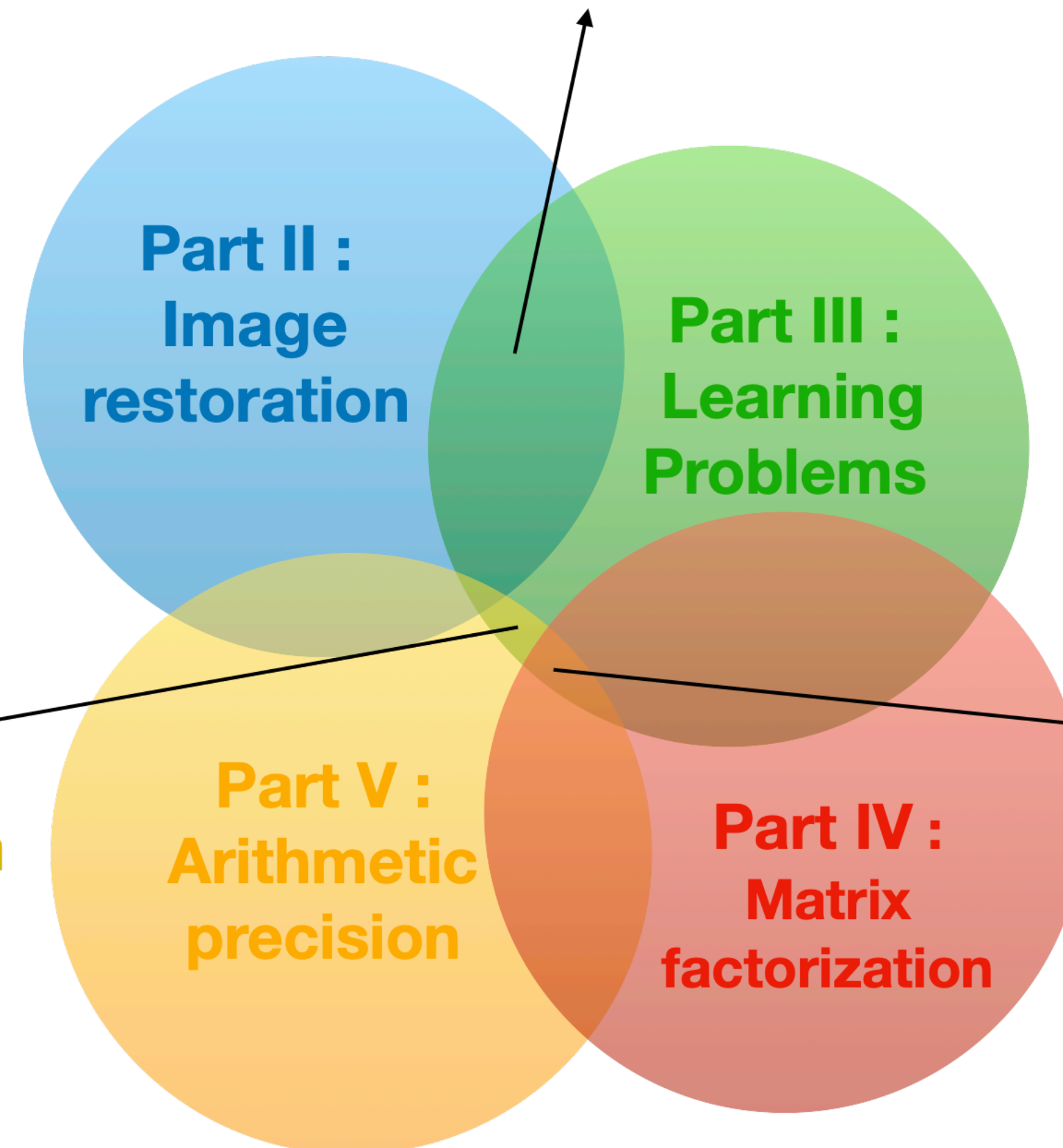


# Part I : multilevel paradigm

- Higher-order order/  
**preconditioned** non-smooth  
ML optimization

- Convergence of the **iterates**  
for ML methods

- Multiresolution neural-network based methods



- Rescaling invariant  
**quantization** of  
neural networks

- Mixed precision **stochastic** optimization
- Mixed precision **SciML**